

8.333 Recitation 1

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References on Probability Theory

For little/no background: Blitzstein and Hwang.

For rigor: Durrett.

Concise summary of everything needed in 8.333:
Chapter 2 of Kardar (Stat Phys of Particles)

Probability Spaces and Measures

A probability space is a triple:

$$(\Omega, \mathcal{F}, \mathbb{P})$$

sample space event space probability measure

Ω : The Sample Space: Space of all possible realizations of experiment.

e.g. for a dice: $\Omega = \{1, 2, \dots, 6\}$.

e.g. throwing a dart: $\Omega =$ a disk in $\mathbb{R}^2$.

$\mathcal{F}$: The Space of Events: a set of subsets of Ω

that satisfies:

define a " σ -algebra" $\left\{ \begin{array}{l} (i) \text{ if } A \in \mathcal{F}, \text{ then } A^c \in \mathcal{F} \\ (ii) \text{ if } A, B \in \mathcal{F}, \text{ then } A \cup B \in \mathcal{F} \end{array} \right.$

We in fact demand the stronger condition $\bigcup_i A_i \in \mathcal{F}$.

for $\{A_i\}$ a countable sequence in $\mathcal{F}$.

e.g. for a dice, all subsets of Ω .

for a dart board, open and closed subsets and unions thereof.

A pair Ω and $\mathcal{F}$ satisfying the above define a measurable space, a space which we can equip with a measure.

A measure is a generalization of the idea of length, area, or volume in $\mathbb{R}^n$.

It is a set function $\mu: \mathcal{F} \rightarrow \mathbb{R}$, satisfying

(i) $\mu(A) \geq 0$, $\mu(\emptyset) = 0$. $\forall A \in \mathcal{F}$

(ii) $\mu(A \cup B) = \mu(A) + \mu(B)$, for disjoint $A, B \in \mathcal{F}$

Demand stronger condition: $\mu(\bigcup_i A_i) = \sum_i \mu(A_i)$

for $\{A_i\}$ a disjoint countable sequence $\subseteq \mathcal{F}$.

e.g. Counting measure μ_c .

For discrete Ω , $\mathcal{F}$ the power set of Ω ,
define $\mu_c(A) = |A|$, where $| \cdot |$ denotes cardinality.

e.g. Lebesgue measure μ_L

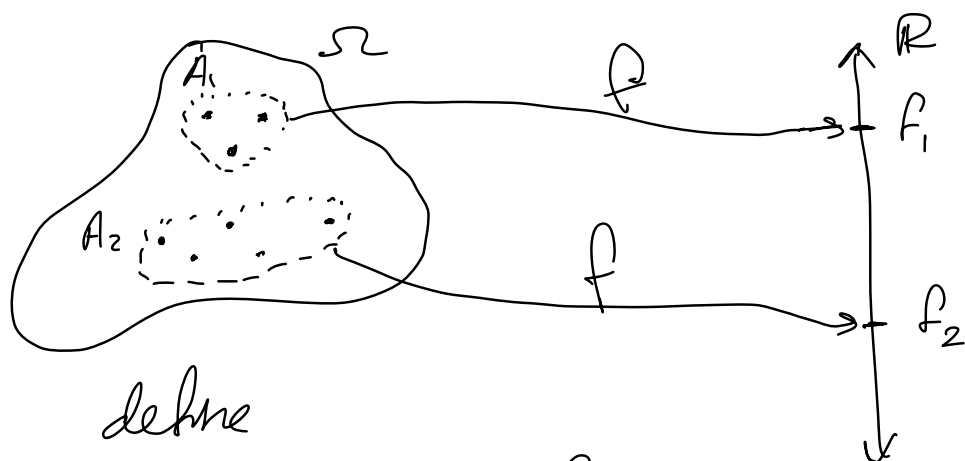
For $\Omega = \mathbb{R}^n$, $\mathcal{B}$ a σ -field on $\mathbb{R}^n$ (the "Borel" sets),
define $\mu_L(A) \equiv$ volume of A .

e.g. for $\Omega = \mathbb{R}$, $\mu_L([a, b]) \equiv b - a$.

Measures can be integrated against.

Formal construction is cumbersome, idea is as follows:

Let $f: \Omega \rightarrow \mathbb{R}$ be a function taking a finite # of values:



Then define

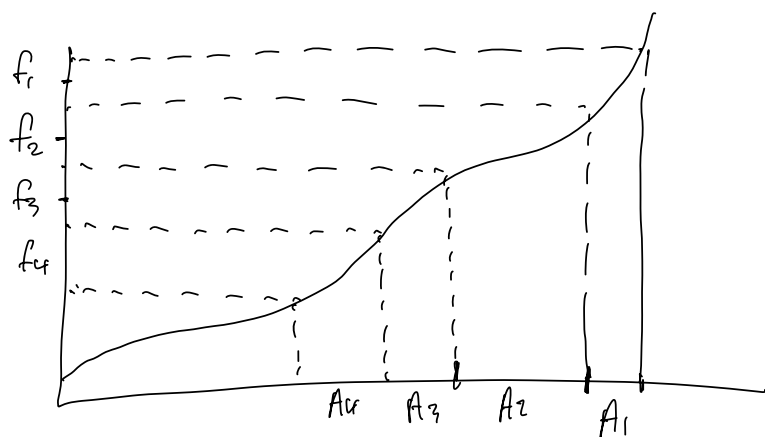
$$\int f d\mu = \sum_i f_i \mu(A_i)$$

Sum over the range of f , weighted by the measure of the pre-images of f . Prob
↑

If f is defined on a discrete set, $f(\omega) = f_\omega$, then the integral w.r.t. counting measure is just the sum

$$\sum_{\omega \in \Omega} f_\omega \mu(\{\omega\}) = \sum_{\omega \in \Omega} f_\omega \quad \text{check at home}$$

For continuous range of f , use a limit



$$\int f d\mu = \lim_{\Delta f \rightarrow 0} \sum f_i \mu(A_i)$$

If μ is Lebesgue measure, then this is just like a Riemann integral but sideways. In fact, can show they are equivalent! " $d\mu_L = dx$ "

A probability measure is a measure satisfying $\mu(\Omega)=1$. I will denote these as $\mathbb{P}$, with $\mathbb{P}(A) \equiv$ probability of event $A \in \mathcal{F}$.

Random Variables

A random variable X is a function $X: \Omega \rightarrow \mathbb{R}$.
Function must be measurable: Roughly, this means the pre-image of any interval in $\mathbb{R}$ is an event
 $X^{-1}(A) \in \mathcal{F}$

This means that statements like " $X \in [0, 1]$ " are events, and can be assigned probabilities under $\mathbb{P}$:
 $\mathbb{P}[X \in [0, 1]] \equiv \mathbb{P}[X^{-1}([0, 1])]$.

An expectation value is the integral of a random variable w.r.t the probability measure:

$$\mathbb{E}(X) \equiv \langle X \rangle \equiv \int X d\mathbb{P}$$

i.e. it is a sum over the outputs of X weighted by the probabilities of the inputs.

$\Rightarrow$ The cumulative distribution function (CDF):

The CDF, $F(x)$, is the probability that X is less than x :

$$F(x) \equiv \mathbb{P}[X \leq x] = \mathbb{P}[X^{-1}((-\infty, x])]$$

$$F: \mathbb{R} \rightarrow [0, 1].$$

$\Rightarrow$ The probability density

For continuous Ω , define $p_x(x) \equiv \frac{dF}{dx}$, where it exists.

Notice:

$$\begin{aligned} \mathbb{P}[X \in [a, b]] &= \mathbb{P}[X \leq b] - \mathbb{P}[X < a] = F(b) - F(a) \\ &= \int_a^b dx p(x). \end{aligned}$$

So $p(x)$ can be interpreted as " $\mathbb{P}[X \in [x, x+dx]]$ ".

For discrete Ω , the probability density $p: X(\Omega) \rightarrow [0, 1]$ is just the probability

$$P_x(x) = \mathbb{P}[X=x].$$

Sometimes called PMF.

What do these have in common? We can think of them as "Jacobians" that convert the probability measure $\mathbb{P}$ into a measure we know how to integrate with. The density P_μ with respect to a measure μ is defined by:

$$\mathbb{P}[A] = \int_A P_\mu d\mu$$

We can interpret as a derivative or Jacobian: $P_\mu \equiv \frac{d\mathbb{P}}{d\mu}$

For continuous Ω , we use densities w.r.t Lebesgue measure

$$\langle X \rangle = \int X d\mathbb{P} = \int X \frac{d\mathbb{P}}{d\mu_c} d\mu_c = \int x p(x) dx$$

For discrete, use counting measure:

$$\langle X \rangle = \int X d\mathbb{P} = \int X \frac{d\mathbb{P}}{d\mu_c} d\mu_c = \sum_{x \in X(\Omega)} x p(x)$$

These are the familiar definitions of expectation that we will be using in this class.

Change of Variables Formula

Let X be an R.V. with density P_x and f an invertible, differentiable function. Define the R.V.

$$Y = f(X)$$

What is the probability density for Y , $P_Y(y)$?

For discrete X , this is easy:

$$P_Y(y) = \mathbb{P}[Y=y] = \mathbb{P}[X=f^{-1}(y)] = P_X(f^{-1}(y))$$

For continuous X , this is slightly less trivial. We require

$$\begin{aligned} \mathbb{P}[Y \in A] &= \mathbb{P}[f(X) \in A] \\ \int_A p(y) dy &= \int_{f(x) \in A} p(x) dx \quad \leftarrow \text{this defines } p(y). \end{aligned}$$

Make a change of variables on the RHS to $y = f(x)$. This requires a Jacobian $\frac{dx}{dy}$, evaluated at y .

$$\int_{f(x) \in A} p(x) dx = \int_{y \in A} p_x(f^{-1}(y)) \left| \frac{dx}{dy} \right| dy$$

We identify the integrand as $P_y(y)$. Thus the formula for change of variables is

$$P_y(y) = P_x(f^{-1}(y)) \left| \frac{df^{-1}}{dy} \right| = P_x(f^{-1}(y)) \left| \frac{1}{f'(f^{-1}(y))} \right|$$

If f is not 1-to-1, i.e. $\exists$ multiple x_i s.t. $f(x_i) = y$ for some y , then we sum over the pre-image of y :

$$P(y) = \sum_i P_x(x_i) \frac{1}{|f'(x_i)|}$$

For $X, Y \in \mathbb{R}^n$, use appropriate Jacobian $J = \det\left(\frac{\partial f_i}{\partial x_j}\right)$.

Remark Since integrals are invariant under a change of measure, expectation values are invariant.

$$\mathbb{E}_y(Y) = \mathbb{E}_x(f(X)).$$

$\therefore$ the mean of an R.V. is invariant.

The mode is the most probable value.

$$x_{\text{mode}} = \arg \max_x P_x(x)$$

This is generally not invariant under a change of parameterization:

$$f(x_{\text{mode}}) \neq \arg \max_y P_y(y).$$

Moments and Cumulants

Moments

The moments of a random variable are defined by

$$m_n = \langle x^n \rangle$$

These fully define a probability distribution. i.e. if two RV have the same moments for all n , then they have the same distribution. Why? consider the characteristic function:

Characteristic Function

The characteristic function is the expectation value

$$\hat{p}(k) = \langle e^{-ikx} \rangle = \int e^{-ikx} dP$$

This always exists because e^{-ikx} is measurable. If the probability density exists, the $\hat{p}$ is its Fourier transform

$$\hat{p}(k) = \int_{-\infty}^{\infty} dx p(x) e^{-ikx}$$

We also have the inversion formula: $p(x) = \int \frac{dk}{2\pi} \hat{p}(k) e^{ikx}$.

Consider the Taylor expansion:

$$\hat{p}(k) = \sum_{n=0}^{\infty} \frac{(-i)^n}{n!} k^n \int_{-\infty}^{\infty} dx p(x) x^n$$

$$= \sum_{n=0}^{\infty} \frac{(-i)^n k^n}{n!} m_n \quad *$$

Thus the moments are the expansion coefficients of $\hat{p}$:

$$m_n = \frac{1}{(-i)^n} \left. \frac{\partial^n \hat{p}}{\partial k^n} \right|_{k=0} = \langle x^n \rangle$$

Cumulant Generating Function

The cumulant generating function is the log of the characteristic function

$$\psi(x) \equiv \log \hat{p}(k) = \log \langle e^{-ikx} \rangle$$

The expansion coefficients are called cumulants, denoted κ_n or $\langle x^n \rangle_c$.

$$\psi(k) = \sum_n \frac{(-ik)^n}{n!} \kappa_n,$$

$$\kappa_n = \frac{1}{(-i)^n} \frac{\partial^n \psi}{\partial k^n} \Big|_{k=0} \equiv \langle x^n \rangle_c$$

ⁿ
Cumulant
or "connected"
average.

Cumulants are important in field theory (they correspond to connected diagrams).

If you take the log of ψ and expand, you get relations between moments and cumulants:

$$\langle x \rangle_c = \langle x \rangle \quad \text{mean}$$

$$\langle x^2 \rangle_c = \langle x^2 \rangle - \langle x \rangle^2 \quad \text{variance}$$

$$\langle x^3 \rangle_c = \langle x^3 \rangle - 3\langle x^2 \rangle \langle x \rangle + 2\langle x \rangle^3 \quad \text{skew}$$

$$\langle x^4 \rangle_c = \langle x^4 \rangle - 4\langle x^3 \rangle \langle x \rangle - 3\langle x^2 \rangle^2 + 12\langle x^2 \rangle \langle x \rangle^2 - 6\langle x \rangle^4 \quad \text{kurtosis}$$